BIOM 417	Critical Care Equipment	3(3,0)
This course focuses on the interpretation of clinical data outputs, the description of modules integration, and the manipulation of features relating to medical equipment (ME) found in surgical theaters, intensive care units, and emergency departments. Students are required to simulate ME's system function, intervene in case of ME's failure, upgrade available ME(s), and, in a team-oriented project, propose and implement a new design for a high-risk medical technology. **Prerequisite:** BIOM 311.		

BIOM 418	Clinical Laboratory Instruments	3(3,0)
This course covers the clinical laboratory responsible for analyzing patient specimens to provide information that aids in the diagnosis of disease and evaluates the effectiveness of therapy. The major topics covered in this course are spectrophotometry, flame photometers, spectrofluorometry, chemistry analyzers, hematology analyzers, and chromatography columns. **Prerequisite**: BIOM 311.		

BIOM 421	Medical Imaging	3(3,0)
This course focuses on the physical and technical features, as well as the clinical applications, of imaging modalities used in healthcare facilities, including diagnostic tools. Specific topics covered include Radiation; X-ray generators; Interaction; Attenuation; Photoelectric effect; Compton scattering; Filters; Collimators; Grids; Intensifying Screen; Fluoroscopic Image Intensifier; Resolution; Ultrasound Generation; US Scan; Transducers; Modes; Doppler. **Prerequisites**: PHYS 312		

BIOM 440	Computational Neuroscience	3(3,0)
This course introduces basic computational methods for understanding what nervous systems do and for determining how they function. It will explore the computational principles governing various aspects of neural coding, dynamics, learning, and memory. Specific topics cover the representation of information by spiking neurons, the processing of information in neural networks, graph theory, and algorithms for adaptation and learning. MATLAB/Python demonstrations and exercises will be used to deepen understanding of various concepts and methods. **Prerequisites**: BIOM 312; MATH 311.		

BIOM 499	CO-OP Work Experience	3(3,0)
This CO-OP work experience is designed to provide students with an eight-week short-term opportunity in biomedical engineering. Students are encouraged to network with industry professionals in their discipline. A report, poster, and PowerPoint presentation are required. **Prerequisites**: ENGL 217; Senior Standing.		

BIOM 519	Advanced Therapeutic Devices	3(3,0)
This course describes the latest biomedical concepts and technologies used for therapeutic purposes. The major topics covered in this course are hemodialysis, hemofiltration, and hemodiafiltration; ventilators and their modes; respiratory modeling and spirometry parameters; and various advanced therapeutic devices. **Prerequisite**: BIOM 414		

BIOM 522	Advanced Medical Imaging	3(3,0)
This course seeks to provide the student with a basic understanding of the concepts of Advanced Biomedical Imaging systems. It involves working with tomographic modalities that produce 3D slice images. The course focuses on the nature of waves, the physical principles, and the instrumentation of each of the three imaging technologies. It applies to radiation interactions with human tissues and to the fundamentals of slice reconstruction systems. Major applications are in Magnetic Resonance, Computed Tomography, Nuclear Medicine, PET, and SPECT scans. Specific topics covered include Radon, 2DFT, translations, rotations, generations, magnetic moment, spin, relaxations, gradient, superconductivity, sequence, contrast, k-space, resolution, artifacts, scintillation, positron, and single photon emission. **Prerequisite**: BIOM 421		

BIOM 531	Clinical Engineering	3(3,0)
This course presents the mission, objectives, and responsibilities of biomedical engineering in health care delivery institutions. It focuses on possessing in-depth knowledge regarding the effective and efficient usage of hospital technological resources. **Prerequisites**: BIOM 417/ BIOM421.		

BIOM 595A	BE Summative Learning Project 1	1(1,0)
The first phase of a team-oriented, project-based experience that culminates in the creation of an artifact; milestones include: project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the device. **Prerequisites**: ENGL 217.		

BIOM 595B	BE Summative Learning Project 2	3(3,0)
The second phase of a team-oriented, project-based experience that culminates in the creation of an artifact; milestones include: project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the created device. **Prerequisite**: BIOM 595A.		

II. Elective Courses

Students may meet their engineering technical electives requirements from the following selection of elective courses.

BIOM 402	Radio Frequency Applications in Biomedical Engineering	3(3,0)
This course addresses radio-frequency design and applications in biomedical engineering and medicine, covering electromagnetic radiation, circuit elements, coupling devices, modulation and demodulation, transmitters, and receivers. Applications include telemetry, transcutaneous power transfer, hyperthermia, RF ablation, tagging, and identification. **Prerequisite**: CCEE 331.		

BIOM 498	Special Topics in Biomedical Engineering	3(3,0)
This course is designed to enable students to study current special topics of interest, carefully selected from BIOM-related areas. The contents of such a course are to be determined by the instructor and the department.		

BIOM 501	Emerging Healthcare Technologies	3(3,0)
This course aims to enable students to review the latest innovations in research areas of interest within the Bio-Engineering field. The course will mainly focus on introducing students to the evolving domain of biomedical technology by examining the interaction and integration of skills among physicians, scientists, engineers, firms, and users. For instance, it will include explicit discussions of fast-growing health care practices that exhibit radical novelty. Students will learn to analyze how such evolutionary technologies can impact a wide range of sectors and associate with several stakeholders. The studies will emphasize topics such as endoscopy, brain mapping, heat therapy, and advanced vital-signs measurements.		

BIOM 502	AI Applications in Healthcare	3(3,0)
This course introduces artificial intelligence (AI) and its applications in healthcare, including machine learning and precision medicine. Students learn the core skills needed to assess clinical information datasets and apply them to enhance evidence-based healthcare outcomes. In this course, students apply AI knowledge and skills to promote effective disease management and patient engagement models.		

BIOM 504	Bio-Image Processing	3(3,0)
This course introduces the basic theories and methodologies of digital image processing. Topics include intensity transformations for image enhancement, the two-dimensional discrete Fourier transform, spatial- and frequency-domain linear image filtering, nonlinear image filtering, binary image processing, edge detection, image segmentation, and the basics of digital video processing. This course makes extensive use of MATLAB as an analysis, design, and visualization tool. **Prerequisite**: CCEE 321.		

BIOM 505	Computer Simulations for Life Sciences	3(3,0)
This course introduces mathematical tools and algorithms implemented to simulate dynamic biological and medical processes. The course examines modeling methodologies and simulation techniques for existing software such as MATLAB/Simulink, LabVIEW, and others. Specific topics covered include System's mode, Compartments, Physiological mechanisms, monitoring medical parameters, Simulation tools, Parameter estimation, Identification and prediction.		

BIOM 506	Biomaterials	3(3,0)
The course studies cell-surface interactions mediated by adhesive proteins in the physiological medium, the analysis of signal transduction, and cellular functions induced by surface control. Emphasis is placed on biomaterial polymers, ceramics, and bone substitutes. Research applications include aiding in the diagnosis of		

malignant tumors and the use of nanobiotechnology to coat medical devices. **Prerequisite**: BIOM 311.

BIOM 507	**Bio-fluids**	**3(3,0)**
This course is designed to enable students to review the main fluids of the human body, their behavior, and the latest related clinical and surgical applications. The course will mainly provide an overview of bodily fluids, basic fluid mechanics concepts, the functional properties of blood, and gas exchange and transport. Also, it includes in-depth discussions of recent health care practices based on the control aspects of the vascular system and the hemodynamics of large arteries.		

BIOM 508	**Microcontroller and Embedded Systems for Biomedicine**	**3(3,0)**
This course explains the architecture of microcontroller-based systems: registers, timers, interrupts, ports, memory, serial communication, and design. It also covers parallel communication and design, analog-to-digital conversion, and the use of PIC 16F84 and PIC 16F877 microcontrollers. MPLAB, PROTEUS, and mikroC software are used. **Prerequisite**: CCEE 426.		

BIOM 512	**Biomechanics and Rehabilitation Engineering**	**3(3,0)**
This course will focus on the human musculoskeletal system. It emphasizes various engineering principles governing biological materials and applies these to the study of bone, cartilage, ligaments, tendons, and muscles. It covers: Bone, Joint, Ligament, Tendon, Cartilage, Viscosity, Elasticity, Plasticity, Stress, Strain, Muscle, Length, Velocity, Moment Arm, Contractile Element, EMG, Lagrangian, and Rehabilitation Mechanics. **Prerequisites**: CIVE 211 and BIOM 311.		

BIOM 513	Hemodynamics Measurement	**3(3,0)**
This course enables the students to study the basic principles of hemodynamics measurement. The students will learn different techniques for measuring blood flow, pressure, compliance, and resistance using invasive and non-invasive methods. In addition, there will be coverage of the qualitative and quantitative measures of cardiopulmonary functions.		

BIOM 521	**Introduction to E-Health Care**	**3(3,0)**
This course emphasizes current applications of telecommunications in medical diagnosis and patient care. It examines collaborative arrangements for real-time management of patients at a distance using wireless, mobile, and robotic technologies. **Prerequisite**: BIOM 421.		

BIOM 532	**Biomedical Servicing**	**3(3,0)**
This course provides students with the technical knowledge and skills necessary to become a broad-based electro-medical specialist. Students will learn to install, set up, troubleshoot, integrate, program, test, operate, and repair systems and components. The topics covered in this course include inspection, corrective and		

preventive maintenance, calibration, repair, supply management, best practices, and accreditation of medical equipment.

BIOM533	**Health-care Facility Management**	**3(3,0)**
This course presents procedures followed for medical and health services management. It covers the duties of the administrator and his assistants, as well as workflow between staff of different services and departments. This course emphasizes topics such as credentials, quality control, human resources, finances, recovery, insurance, sterilization, infection control, and health safety procedures.		

BIOM 597	**Advanced Topics in Biomedical Engineering**	**3(3,0)**
This course is designed to enable students to study a specific advanced topic of interest, carefully selected from biomedical engineering-related topics. The content outline for such a topic is to be determined by the instructor and approved by the department Chair. **Prerequisite**: Senior Standing		